Innovations in Pricing

R K Amit

Department of Management Studies
Indian Institute of Technology Madras
Chennai 600 036
✉: rkamit@iitm.ac.in
☎: +91-44-2257 4575

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Demand

Will taxing cigarettes reduces smoking?



Demand



MoviePass's policy changes come after a tumultuous several weeks during which it briefly stopped working in most theaters and parent Helios had to borrow money at a high interest rate to stay afloat. **PHOTO:** DARRON CUMMINGS/ASSOCIATED PRESS

By Ben Fritz 112 COMMENTS

Updated Aug. 6, 2018 6:46 p.m. ET

The subscription service MoviePass sharply scaled back how many new films customers can see for its monthly fee, hoping to stanch the financial bleeding after a tumultuous few weeks of service outages and withering reviews that have chopped its parent company's share price to pennies.

The company, which a year ago began charging just $9.95 a month to let users see movies as often as once a day at most U.S. theaters, said it would soon begin limiting customers to three a month. That new policy is in lieu of other recently announced changes, including a boost in the monthly subscription fee to $14.95, that were highly unpopular and drove away customers, adding to the company's financial strains.

Demand

- Operations & Marketing
- Is this a reasonable model of what real firms do?
- Demand & Inverse Demand Functions
- Demand Facing a Firm versus Facing an Industry—Aggregation
- Commodities—Perfect substitutes
- Differentiated
- Competitive Firms—*Almost* No Control
- Firms with market power

Demand Functions



Why is it horizontal? Why for some it slope downwards? Upward?

Demand Functions

Do firms know their demand functions?



Amazon.com, Flipkart.com, ...

Elasticity

- The elasticity along the demand function $D(p)$ at a given price is the rate of percentage change in quantity demanded per 1% change in the price, starting from that price.

$$\nu(p_0) = \frac{\Delta x/x_0}{\Delta p/p_0}$$

$$\widehat{\nu}(x_0)?$$

- Computing elasticity at different prices for $D(p) = 1000(20 - p)$.
- Compute elasticity for $D(p) = Cp^k$.
- Why percentage changes?

Marginal Revenue & Elasticity

$$MR(x) = P(x)\left[1 + \frac{1}{\widehat{\nu}(x)}\right]$$

- $\widehat{\nu}(x) = -1$, demand has unit elasticity, $MR(x) = 0$.
- $\widehat{\nu}(x) < -1$, demand is elastic, $MR(x) > 0$.
- $\widehat{\nu}(x) > -1$, demand is inelastic, $MR(x) < 0$.

♠ What is the marginal revenue of a firm operating in a perfectly competitive market?

Elasticity of Demand for Crude Oil

Category	Estimated $\epsilon_{Q,P}$
Airline travel, leisure	−1.52
Rail travel, leisure	−1.40
Airline travel, business	−1.15
Rail travel, business	−0.70
Urban transit	between −0.04 and −0.34

Source: Elasticities from the cross-sectional studies summarized in Tables 2, 3, 4 in Tae Hoon Oum, W. G. Waters II, and Jong-Say Yong, "Concepts of Price Elasticities of Transport Demand and Recent Empirical Estimates," *Journal of Transport Economics and Policy* (May 1992): 139–154.

Elasticity

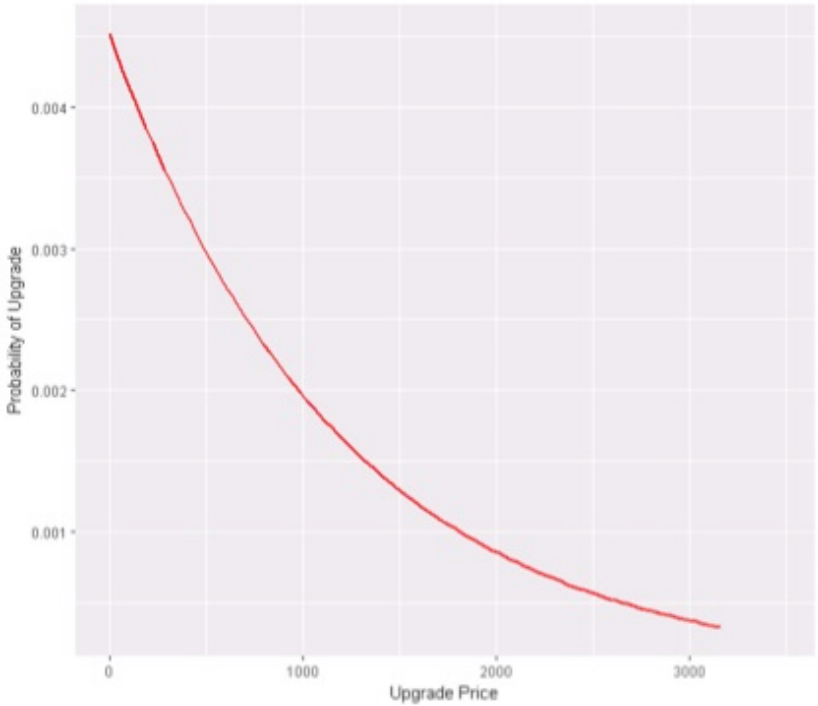
	Price Elasticity	
Country	**Short-Run**	**Long-Run**
Australia	−0.034	−0.068
France	−0.069	−0.568
Germany	−0.024	−0.279
Japan	−0.071	−0.357
Korea	−0.094	−0.178
Netherlands	−0.057	−0.244
Spain	−0.087	−0.146
United Kingdom	−0.068	−0.182
United States	−0.061	−0.453

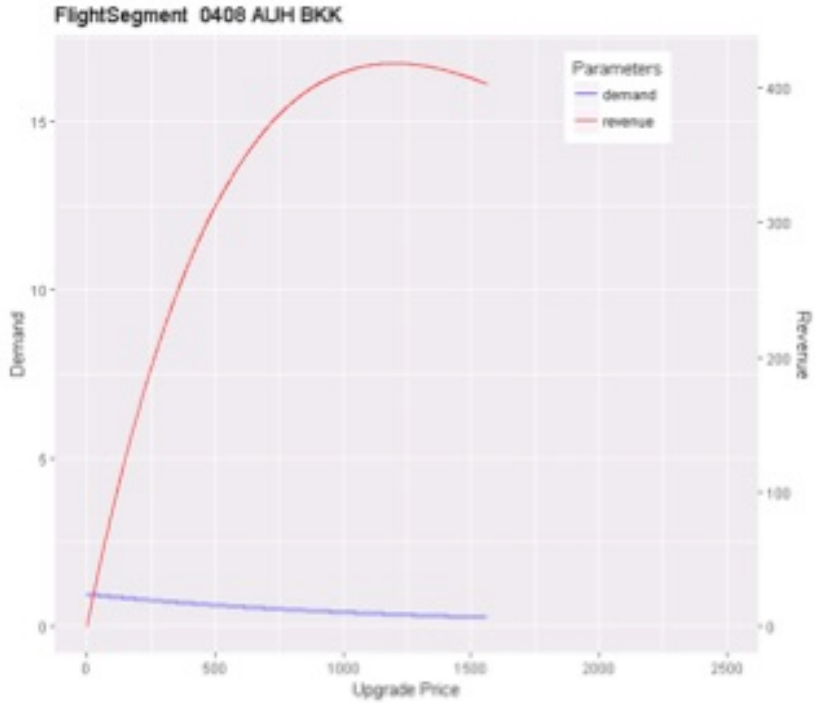
Marginal Revenue & Elasticity

$$P(x) = 20 - 2x$$

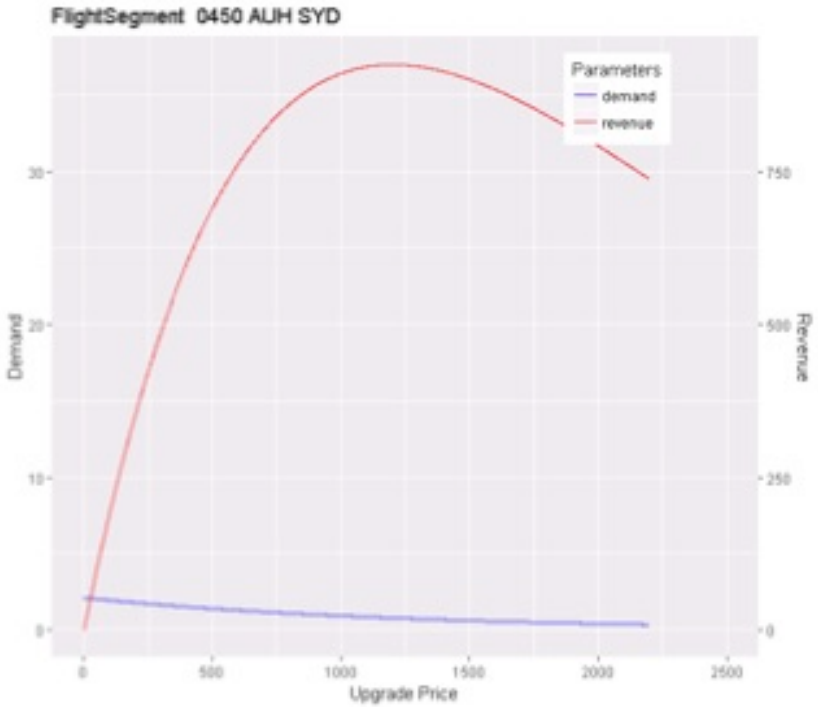
Quantity	Price	Elasticity	MR	Revenue
0	20	-∞	20	0
1	18	-9.00	16	18
2	16	-4.00	12	32
3	14	-2.33	8	42
4	12	-1.50	4	48
5	10	-1.00	0	50
6	8	-0.67	-4	48
7	6	-0.43	-8	42
8	4	-0.25	-12	32
9	2	-0.11	-16	18

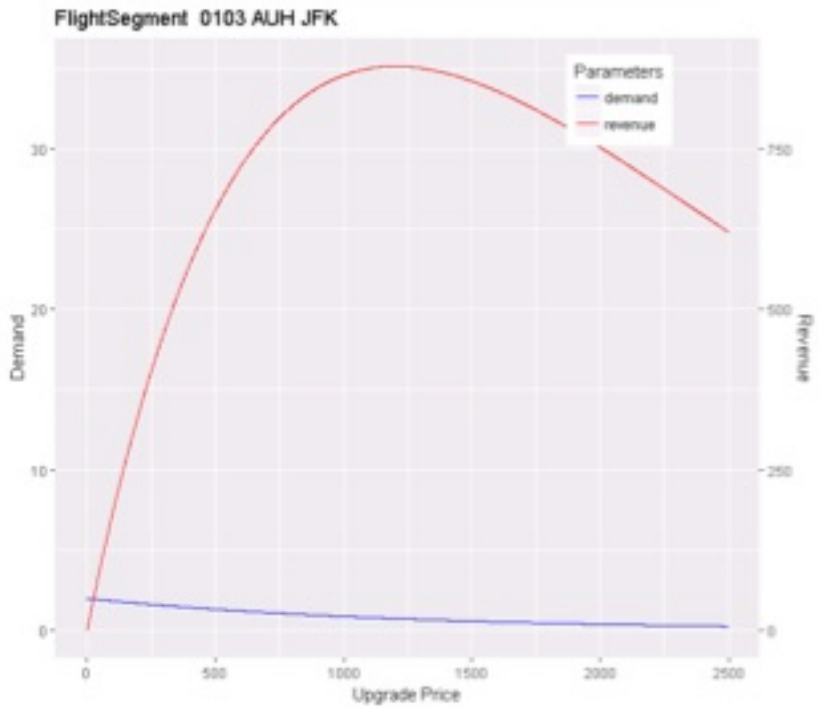
Pricing for Upgrading





Pricing for Upgrading





What makes demand more elastic?

- Products with close substitutes have more elastic demand.
- Demand for an individual brand is more elastic than industry aggregate demand.
- Products with many complements have less elastic demand.
- In the long run, demand curves become more elastic.
- As price increases, demand becomes more elastic.

♣ No profit-maximizing firm with market power and positive marginal costs should produce at a level where its demand is inelastic.

Motivation for Complex Pricing

- In many corporations, the selection of prices used to be something of an afterthought.
- In the past 20 years, pricing has grown in most organizations as a major strategic concern.
- American Airlines estimates its yield management generates $500 million per year in additional revenue. They excel in Price Discrimination—charging different customers different prices for the (nearly) same good.
- Cost-based versus Value-based pricing. A monopoly can charge $P(x)$—same price for all the units for each customer. Can it do better?



Some Examples

- International pricing by the pharmaceutical companies
- Methyl Methacrylate—a plastic used extensively in the manufacture of dentures. DuPont and Rohm & Haas price the powdered version at \$0.85 per pound for industrial uses; \$22 per pound for the liquid MM for the dental labs. Rohm & Haas planted a rumor they had adulterated the powered MM.
- Niagara Mohawk Power Corporation—an electric utility's individualized pricing to induce them not to generate their own electricity.
- Prior to 2007, Jet Airways, Indian Airlines, and Air Sahara charged foreigners in US dollars; these dollar prices were generally priced about 50% more than the rupee fare.

Some Examples

ECONOMIC VIEW

Tesla's Tiered Pricing Is a Hurdle, but a Fair One



A Tesla Model 3 sedan pictured in a company handout. The car, whose price starts at $35,000 before incentives, is meant to help the company achieve greater scale. Tesla, via Reuters

By Robert H. Frank

Oct. 27, 2017

Type of Price Discrimination

Caveats Transfer of the commodity; gray markets. Carrots and Sticks to differentiate the product.

First Degree PD The ideal form of PD. The firm identifies the valuation of each buyer and prices accordingly. Unlikely to exist in practice.

Second Degree PD This identifies imperfectly the valuation of the buyers thorugh a form of self-selection. For example, coupons in the Sunday supplements, business and economy class tickets, etc. Also called Indirect PD.

Third Degree PD This involves the use of a customer signal—age, occupation, usage, income, race, etc.) to discriminate. Examples include student discounts. Also called Direct PD.



Third Degree PD

- Charge higher-valuation customers more than lower-valuation customers for the same product. This is done using some observable signal.
- Software prices are lower for schools and universities as compared with corporate buyers. Magazine subscription rates are different for the students.
- The monopoly pricing rule states that

$$P^*(x)\left[1+\frac{1}{\widehat{\nu}(x)}\right]=MC(x)$$

If $\widehat{\nu}(x)=-2$, $P^*(x)=2MC(x)$. This is 100% markup.

Third Degree PD

Lerner Index of Market Power

$$\frac{P - MC}{P}$$

It ranges from 0 to 1. It is zero for a perfectly competitive market.

Third Degree PD

- UM & RKA, 2015 observe high markups in vegetables and fruits retailing in Chennai. These markups vary with locations for the same retailer—marginal costs are approximately same for all the locations.
- GM and Ford give discounts to the first-time buyers. Oracle sells essentially the same enterprise software at a substantially lower price to midmarket customers compared with large-enterprise customers.
- Kevlar, a fibre by DuPont, is lighter and stronger than steel. It is used in bulletproof vests, racing shoes, and tyres. DuPont is able to charge different prices for Kevlar depending on the end-use applications.

Second Degree PD

- The seller offers a menu of choices to the buyer, from which the buyer selects. The trick is to encourage with higher valuation to choose the expensive items from the menu.
- Carrots and Sticks—Dupuit (1844) notes about wooden benches in the third class coaches

  [W]hat the company is trying to do is to prevent the passengers who can pay the second class fare from traveling third class; it hits the poor not because it wants to hurt them, but to frighten the rich.

Second Degree PD: Versioning

- Two most prevalent ideas in second degree PD: Versioning and Bundling.
- Versioning—add/subtract features of the product in different ways to offer multiple versions of the same basic product.
- Cannibalization—For example, buy-down behavior in airlines.
- Example: Two versions of the printer. Two customer segments with unit demand in each. The valuations are

	Fast	Slow
v_A	\$100	\$40
v_B	\$49	\$30

- Can you suggest the pricing?
- Different versions of the software; Hardback versus paperback books; Priceline.com

Second Degree PD: Bundling

- A bundle is a collection of distinct products or services that are sold together as a package.
- Microsoft Office; Vacation packages; Lunch buffet; Cable TV.

Pure Bundling/Tying The seller combines two or more products into one product. The individual products are not for sale. Until 2010, the Windows OS could not be purchased without Internet Explorer.

Mixed Bundling The seller gives the buyer the option of buying the bundle at a discount, or any subset of the bundle. It can be used even when the products are unrelated. AT&T used bundling long-distance phone services and coupons for oil changes at Jiffy Lube.



Second Degree PD: Bundling

- Suggest—no bundling/pure bundling/mixed bundling.

	Indian Ocean	Zakir Hussain
v_A	\$80	\$40
v_B	\$40	\$80
v_C	\$90	\$10
v_D	\$10	\$90

- Revenue in no bundling=\$320. Pure bundling=\$400?. Mixed bundling = \$420.
- In the pure bundling case, the variance for the bundle is lower, which helps the seller.
- Nonlinear pricing—bundling multiple units of the same item.

Second Degree PD: Bundling

"The key intuition behind the power of bundling is that consumer's valuation for a collection of goods typically has a probability distribution with a lower variance per good compared to the valuations for the individual goods. The larger the number of goods bundled, the greater the typical reduction in the variance. Under a very general set of conditions, the law of large numbers guarantees that the distribution of valuations for the bundle has proportionately fewer extreme values."—Yannis Bakos & Erik Brynjolfsson, "Bundling Information Goods: Pricing, Profits, and Efficiency", Management Science, 1999.

"[s]uch a reduction in *buyer diversity* typically helps sellers extract higher profits from all consumers."—Richard Schmalensee, "Gaussian demand and commodity bundling", Journal of Business, 1984.



Revenue or Yield Management

- Products like cruise ships, hotels, airlines have several characteristics that affect their pricing
- Assume $MR > MC$, a monopolist reduce prices to sell more, but cannot due to capacity constraints. In such cases, the monopolist prices to match demand with supply.
- If the demand is known, this is relatively easy to do. However, in most cases, the demand is uncertain.
- The manager will tradeoff the costs of overpricing—lost profit on unsold items, and the costs of underpricing—lower margins on all items.

Revenue or Yield Management

- Revenue increases when the monopolist offer two prices for the same product, and limit the quantity sold at the lower price.
- Reserving a number of items at the low price is effectively an auction. For example, the airlines allows one to buy tickets at low price by forgoing flexibility of being able to change dates. They offer advance purchase discount to induce low-price customers to arrive first.



Revenue or Yield Management: An Example

- Assume ITC Chola has inventory of 100 rooms for Oct 31, 2026. What price do you post?
- Trade-offs?
- Assume that the future demand for hotel rooms will be in one of two states—high (**H**) and low (**L**) with equal probability. In **H** there will be 100 buyers available each with $v = \$120$; in **L** there will be 50 buyers with $v = \$100$.
- The state is unknown. Compute the revenue for a fixed posted price.
- P= 100; $\mathbb{E}$R= $\$7,500$.
- P= 120; $\mathbb{E}$R= $\$6,000$.
- Can they do better?

Revenue or Yield Management: An Example

- Price 50 rooms at \$100 and 50 rooms at \$120. In state **L**, the revenue will be \$5000. In state **H**, the revenue will be \$11,000. $\mathbb{E}$R= $\$8,000$.
- Revenue increases when you offer two prices for the same product and limit the quantity sold at the lower price.
- Determining the number of price levels and number of units for each level called protection levels is a complex optimization problem.
- A unit sold today at a low price could have been sold tomorrow at a higher price. This trade-off is absent when the supply is unlimited.

What are Auctions?

- Auctions are highly structured market transactions primarily used in thin markets—markets with few participants and infrequent transactions.
- Auctions are best for periodic or episodic high-volume and high-value transactions.
- An auction is defined by
  1. Bidding format rules
  2. Bidding process rules
  3. Price and allocation rules

Types of Auctions

- Auctions have been around for centuries—at least since the time of *Herodotus* in 500 BCE.
- Most price-only auctions used some variant on one of four basic forms
  1. English, ascending price
  2. Dutch, descending price
  3. First-price sealed bid
  4. Second-price sealed bid
- Auctions also differ in how many units are transacted and how often—single object, multi-object, recurring auctions, etc.
- Forward and Reverse Auctions



A New Age of Auctions

- Auctions have changed. One reason is game theory.
- Game theory provides mathematical tools and techniques to analyze alternative auction designs.
- Auction design in telecommunications, energy, and natural resources. Google, Microsoft, Yahoo, eBay, are pursuing research in auction design.
- Multi-object auctions—Simultaneous multi-round auction, simultaneous ascending auction, simultaneous ascending and descending clock auctions.

Auctions as Mechanisms

- The roots of mechanism design/implementation theory lie in the Hayek-Mises-Lange-Lerner debates on the feasibility of market socialism.
- In the 1950s and 1960s, Leonid Hurwicz extended them to other mechanisms.
- Mechanism as being unknown—the economic institution by which the economic activity is to be coordinated. Economic environment as the parameters.
- Are there alternative mechanisms to the price system which would lead not only to efficient but also to equitable outcomes?



Auctions as Mechanism Design

- Direct Mechanism—A mechanism in which the equilibrium bids submitted by players are always equal to their valuations. Indirect Mechanism.
- FPSB auction is an example of an indirect mechanism. SPSB auction is an example of a direct mechanism.
- Revelation Principle: *If there exists any mechanism that implements a social choice function in dominant strategies then there exists a direct mechanism that implements in dominant strategies and is truthful.*
- The revelation principle restricts our search of optimal auctions to direct mechanisms.



First-Price and Dutch

Theorem 1.

First-Price and Dutch auctions are strategically equivalent.

- In both first-price and Dutch, a bidder must decide on the amount he's willing to pay, conditional on having placed the highest bid.
  - despite the fact that Dutch auctions are extensive-form games, the only thing a winning bidder knows about the others is that all of them have decided on lower bids
    - e.g., he does not know *what* these bids are
    - this is exactly the thing that a bidder in a first-price auction assumes when placing his bid anyway.
- Note that this is a stronger result than the connection between second-price and English.



Discussion

- So, why are both auction types held in practice?
  - First-price auctions can be held asynchronously
  - Dutch auctions are fast, and require minimal communication: only one bit needs to be transmitted from the bidders to the auctioneer.
- How should bidders bid in these auctions?
  - They should clearly bid less than their valuations.
  - There's a tradeoff between:
    - probability of winning
    - amount paid upon winning
  - Bidders don't have a dominant strategy any more.



FPSB Auction

Theorem 2.

In a first-price auction with two risk-neutral bidders whose valuations are drawn independently and uniformly at random from $[0, 1]$, $(\frac{1}{2}v_1, \frac{1}{2}v_2)$ *is a Bayes-Nash equilibrium strategy profile.*

Proof.

Assume that bidder 2 bids $\frac{1}{2}v_2$, and bidder 1 bids b_1. From the fact that v_2 was drawn from a uniform distribution, Assume that bidder 2 bids $\frac{1}{2}v_2$, and bidder 1 bids s_1.

From the fact that v_2 was drawn from a uniform distribution, hence bidder 2 bids are between 0 and $\frac{1}{2}$. Bidder 1, with valuation v_1, bids $b_1 \in \left[0, \frac{1}{2}\right]$ then he wins only if $v_2 < 2b_1$. This gives optimal $b_1^* = \frac{v_1}{2}$. ■



SPSB Auction

Theorem 3.

Truth-telling is a dominant strategy in a second-price auction.

- In fact, we know this already (do you see why?)
- However, we'll look at a simpler, direct proof.

Second-Price proof

Theorem 4.

Truth-telling is a dominant strategy in a second-price auction.

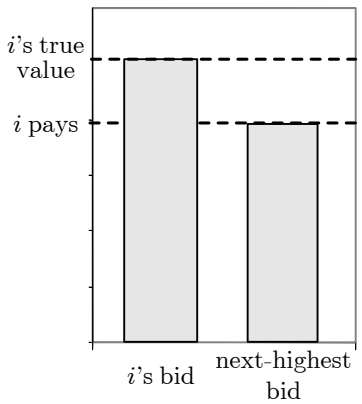
Proof.

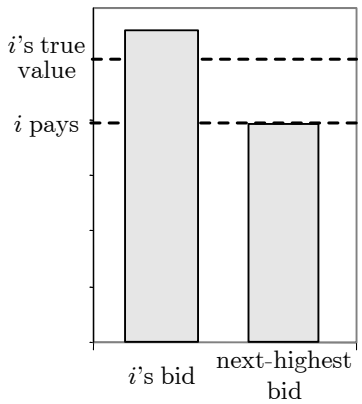
Assume that the other bidders bid in some arbitrary way. We must show that i's best response is always to bid truthfully. We'll break the proof into two cases:

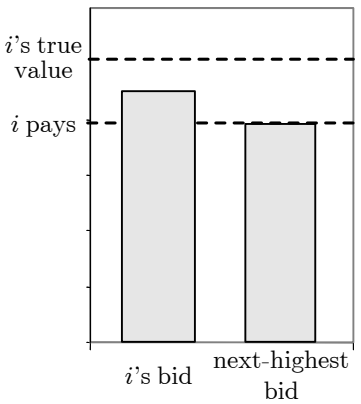
1. Bidding honestly, i would win the auction
2. Bidding honestly, i would lose the auction

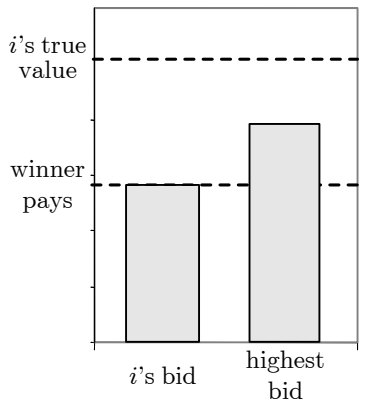
■

Second-Price proof (2)



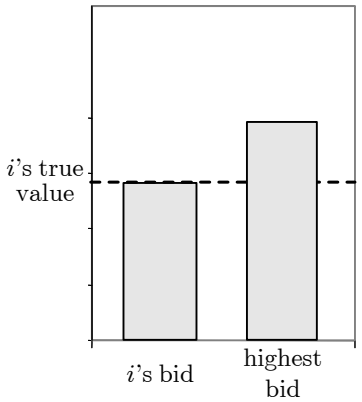


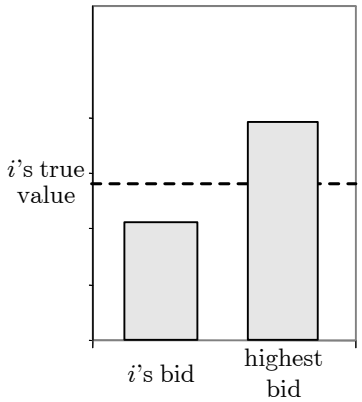


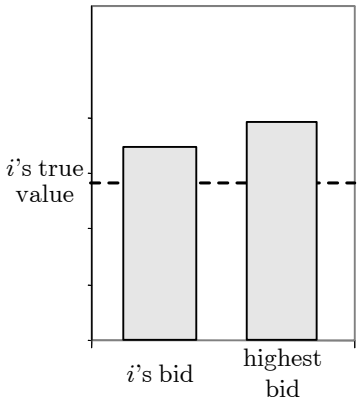


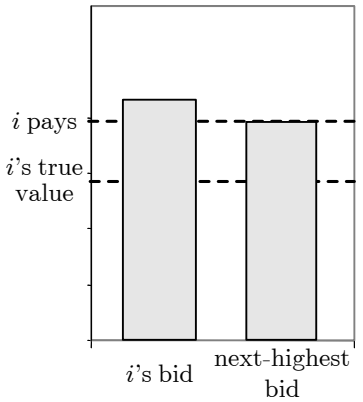
- Bidding honestly, i is the winner
- If i bids higher, he will still win and still pay the same amount
- If i bids lower, he will either still win and still pay the same amount. . . or lose and get utility of zero.

Second-Price proof (3)









- Bidding honestly, i is not the winner
- If i bids lower, he will still lose and still pay nothing
- If i bids higher, he will either still lose and still pay nothing. . . or win and pay more than his valuation.

Revenue Equivalence

- Which auction should an auctioneer choose? To some extent, it doesn't matter...

Theorem 5 (Revenue Equivalence Theorem).

Assume that each of n risk-neutral agents has an independent private valuation for a single good at auction, drawn from a common cumulative distribution $F(v)$ that is strictly increasing and atomless on $[\underline{v}, \bar{v}]$. Then any auction mechanism in which

- *the good will be allocated to the agent with the highest valuation*
- *any agent with valuation $\underline{v}$ has an expected utility of zero*

yields the same expected revenue, and hence results in any bidder with valuation v making the same expected payment.

Revenue Equivalence

- The kth order statistic of a distribution: the expected value of the kth-largest of n draws.
- For n IID draws from $[0, v_{max}]$, the kth order statistic is

$$\frac{n+1-k}{n+1} v_{max}.$$

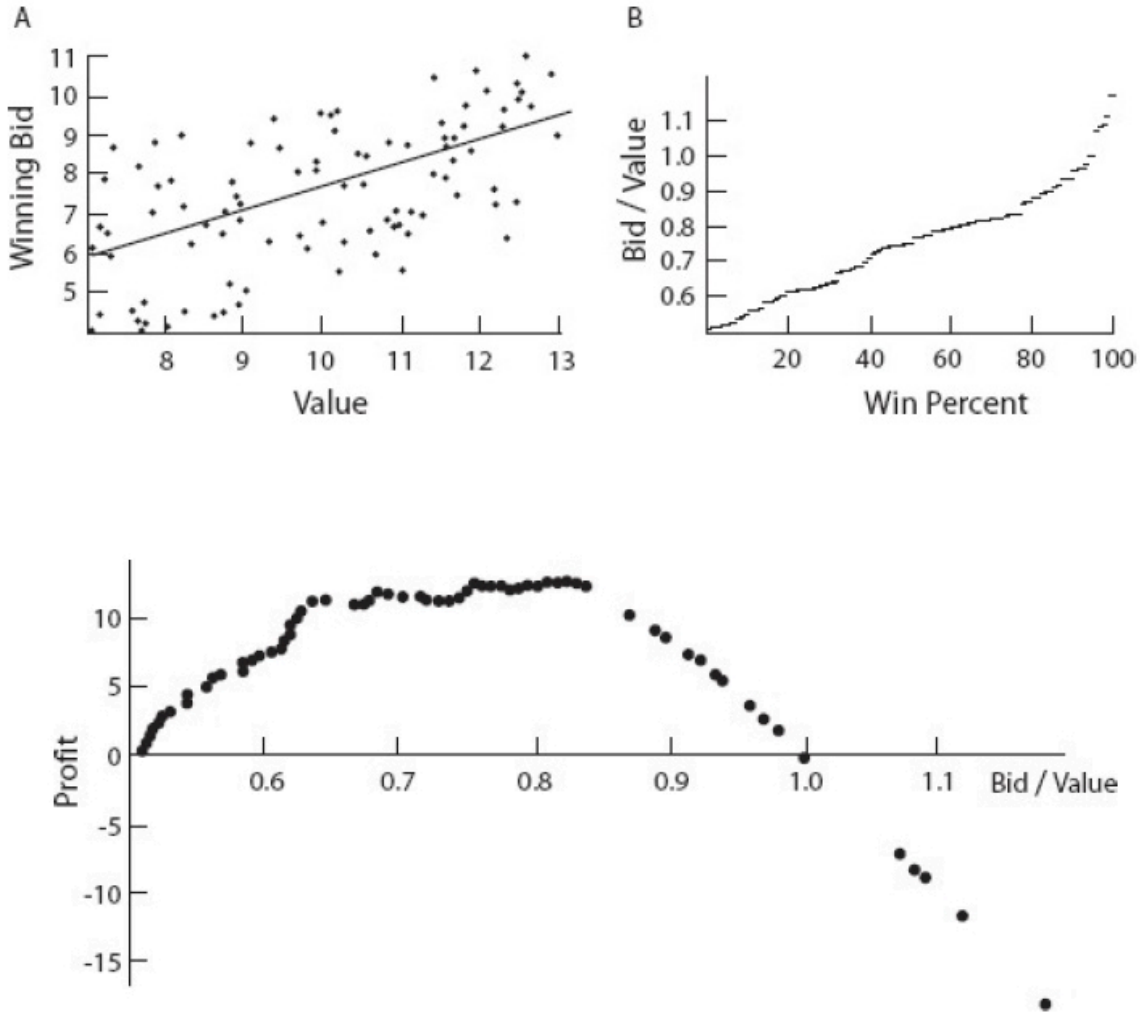
- Thus in a second-price auction, the seller's expected revenue is

$$\frac{n-1}{n+1} v_{max}.$$

- First and second-price auctions satisfy the requirements of the revenue equivalence theorem.



Bid as a Function of Value



Vickrey-Clarke-Groves (VCG) Mechanism

SPSB is a special case of VCG mechanism. The VCG mechanism has an outcome function

$$x^*(\theta) = \arg\max_{x \in X} \sum_i u_i(x, \theta_i) \qquad \forall \theta \in \Theta$$

$$m_i(\hat{\theta}) = \sum_{j \neq i} u_j(x^*(\hat{\theta}), \hat{\theta}_j) - \sum_{j \neq i} u_j(x^*_{-i}(\hat{\theta}_{-i}), \hat{\theta}_j)$$

Theorem 6.

Any VCG mechanism is truthfully implementable in dominant strategies.

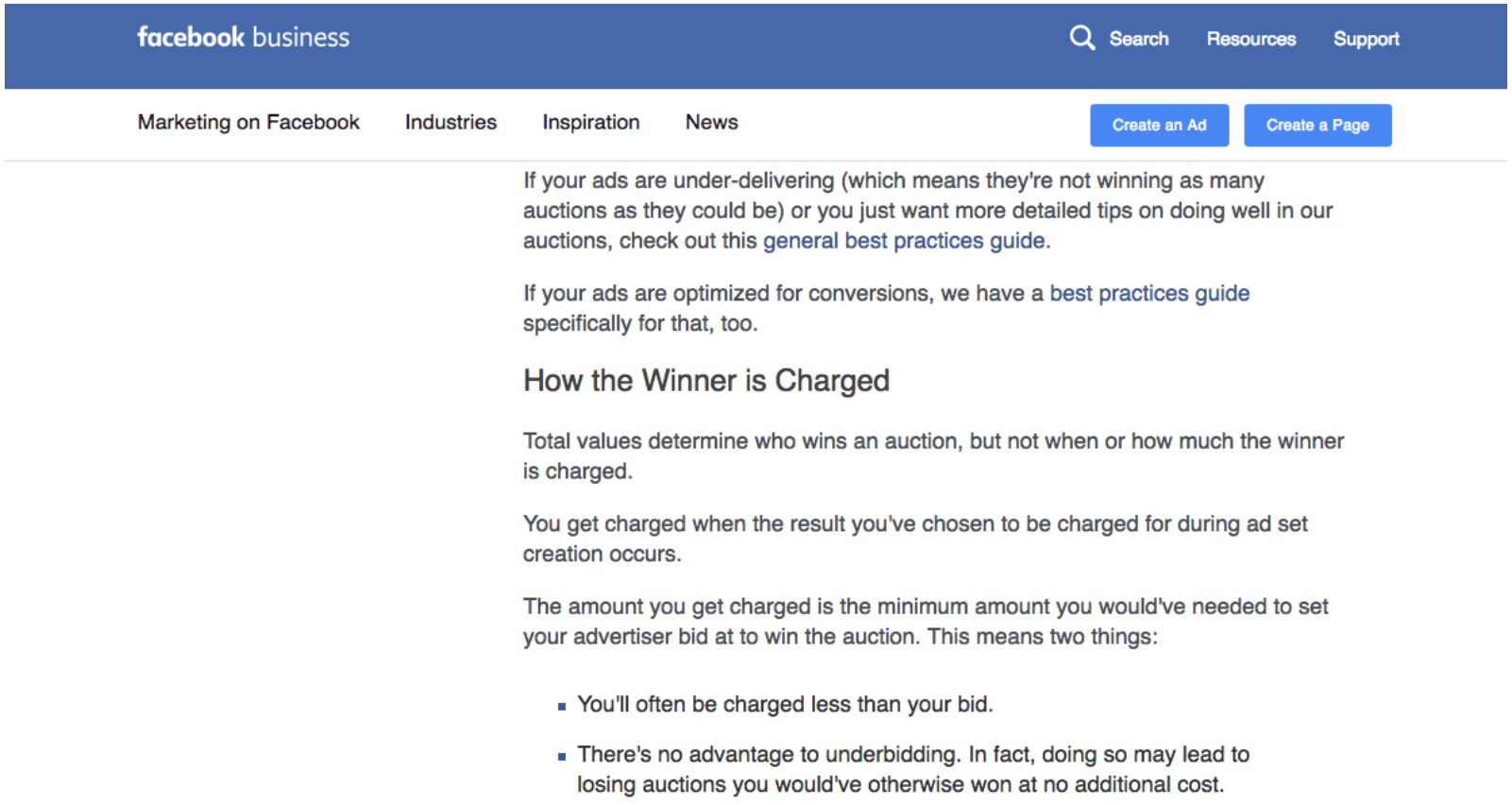
Winner's Curse



Internet Advertising

- In 1994, the first advertisement appeared on the internet—a simple banner ad.
- Early internet advertising was sold on the basis of impressions—a company like Yahoo! would negotiate a rate with an advertiser, agreeing on a price for displaying the ad a fixed number of times.
- This kind of advertising is not related to the behavior of the users. When the dot-com bubble burst in 2000, online advertising started declining.
- At that time, Google introduced the idea of keyword based advertising—Adwords.
- Google understood the importance of isolating the ads from the search results and sorting them in order of relevance.

VCG at Facebook



VCG at Facebook

Facebook uses the **Vickrey-Clarke-Groves (VCG)** auction system to sell and select ads for the network. A more useful explanation what we, as users need to know about this side of VCG came from Facebook chief economist **John Hegeman**, in an interview.

"If you're an advertiser and you're getting a chance to show your ad, you're going to take away the opportunity from someone else. The price [of the ad] can be determined based on how much value is being displaced from those other people. An advertiser will only win this placement if their ad really is the most relevant, if it really is the best ad to show to this person at this point in time."

Google

hotels in chennai

All Maps News Images Videos More Settings Tools

About 2,26,00,000 results (0.99 seconds)

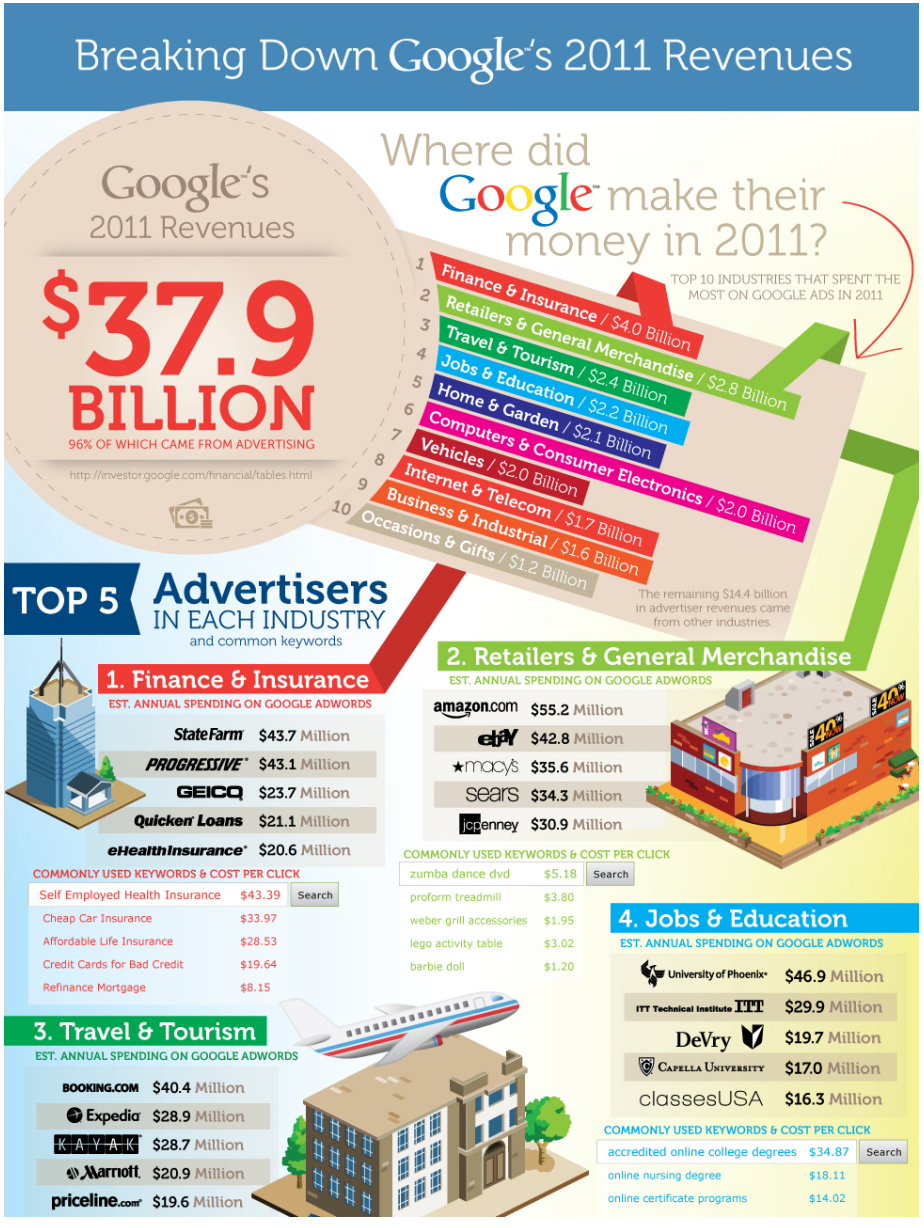
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Amenities: WiFi, Pool, Breakfast, Banquet

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FabHotels Official website. Book Direct & Get Flat 35% Off. Use code: CHN35




Breaking Down Google's 2011 Revenues
Google's 2011 Revenues
$37.9 BILLION
96% OF WHICH CAME FROM ADVERTISING
http://investor.google.com/financial/tables.html
Where did Google make their money in 2011?
TOP 10 INDUSTRIES THAT SPENT THE MOST ON GOOGLE ADS IN 2011
1 Finance & Insurance / $4.0 Billion
2 Retailers & General Merchandise / $2.8 Billion
3 Travel & Tourism / $2.4 Billion
4 Jobs & Education / $2.2 Billion
5 Home & Garden / $2.1 Billion
6 Computers & Consumer Electronics / $2.0 Billion
7 Vehicles / $2.0 Billion
8 Internet & Telecom / $1.7 Billion
9 Business & Industrial / $1.6 Billion
10 Occasions & Gifts / $1.2 Billion
The remaining $14.4 billion in advertiser revenues came from other industries.
TOP 5 Advertisers IN EACH INDUSTRY and common keywords
1. Finance & Insurance
EST. ANNUAL SPENDING ON GOOGLE ADWORDS
StateFarm $43.7 Million
PROGRESSIVE $43.1 Million
GEICO $23.7 Million
Quicken Loans $21.1 Million
eHealthInsurance $20.6 Million
COMMONLY USED KEYWORDS & COST PER CLICK
Self Employed Health Insurance $43.39 Search
Cheap Car Insurance $33.97
Affordable Life Insurance $28.53
Credit Cards for Bad Credit $19.64
Refinance Mortgage $8.15
2. Retailers & General Merchandise
EST. ANNUAL SPENDING ON GOOGLE ADWORDS
amazon.com $55.2 Million
eBay $42.8 Million
macy's $35.6 Million
sears $34.3 Million
jcpenney $30.9 Million
COMMONLY USED KEYWORDS & COST PER CLICK
zumba dance dvd $5.18 Search
proform treadmill $3.80
weber grill accessories $1.95
lego activity table $3.02
barbie doll $1.20
3. Travel & Tourism
EST. ANNUAL SPENDING ON GOOGLE ADWORDS
BOOKING.COM $40.4 Million
Expedia $28.9 Million
KAYAK $28.7 Million
Marriott $20.9 Million
priceline.com $19.6 Million
4. Jobs & Education
EST. ANNUAL SPENDING ON GOOGLE ADWORDS
University of Phoenix $46.9 Million
ITT Technical Institute ITT $29.9 Million
DeVry $19.7 Million
Capella University $17.0 Million
classesUSA $16.3 Million
COMMONLY USED KEYWORDS & COST PER CLICK
accredited online college degrees $34.87 Search
online nursing degree $18.11
online certificate programs $14.02

Two Innovations in Internet Advertising

Pay-Per-Click Advertising In AdWords, the advertisers only pay if a user clicks on their links based on "price-per-click" (PPC). The success rate of an advertisement is measured by its click-through rate (CTR).

Setting Prices through an Auction How a search engine set the prices per click for different queries. One possibility is simply to "post the prices". With so many possible keywords and combination of keywords, this is an extremely difficult task.

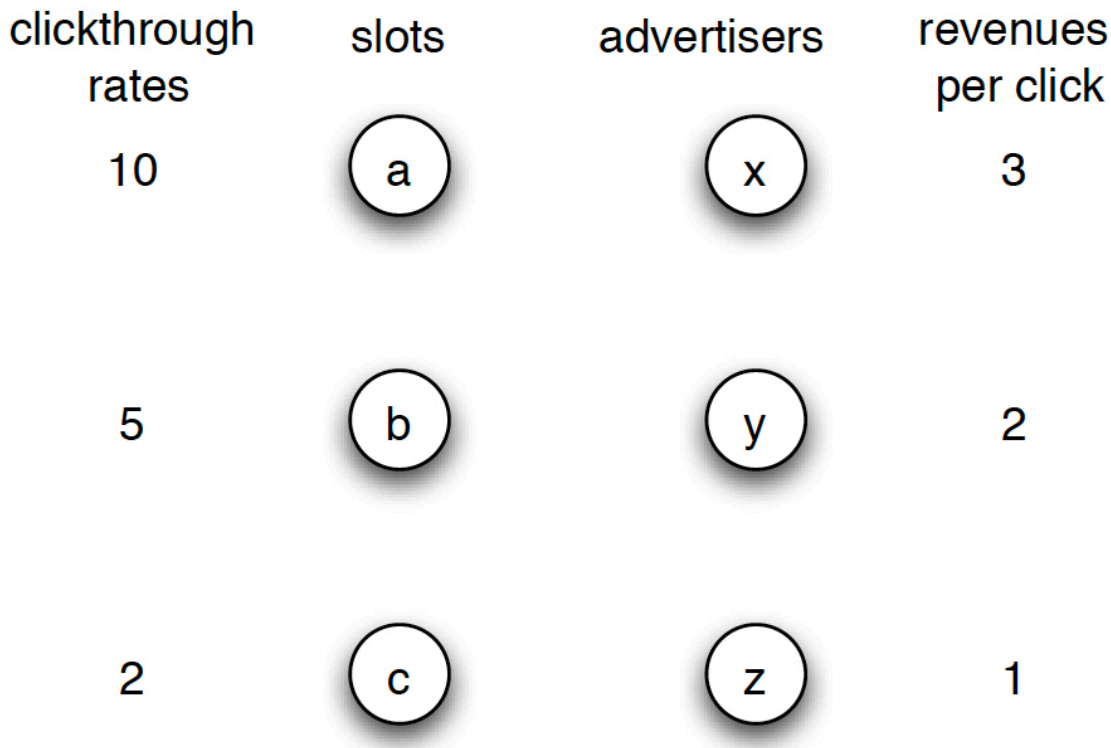
Search engines use auctions to determine prices. The VCG mechanism is a natural choice. However, the auction mechanism that AdWords uses is the generalized second-price auction (GSP).



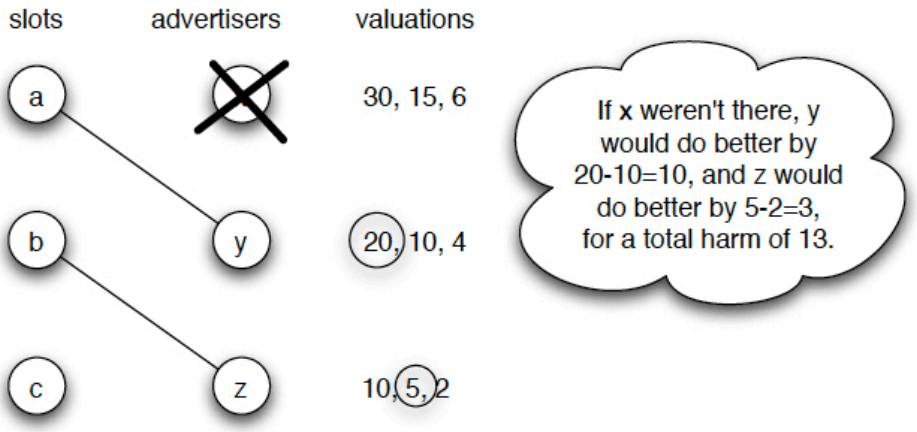
GSP Auction

- A search engine page contains a fixed number of possible positions for advertisements.
- The simplest GSP auction works as follows
  1. Advertiser j, with valuation v_j, submits a bid b_j.
  2. After the bidding, the GSP auction awards each slot i to the ith highest bidder at a price per click equal to b_{i+1}.
  3. For slot i, the expected payment is CTR $\times$ b_{i+1}.
- Although the GSP auction has a simple description, the bidding behavior is *untruthful* and leads to *socially non-optimal outcomes*.

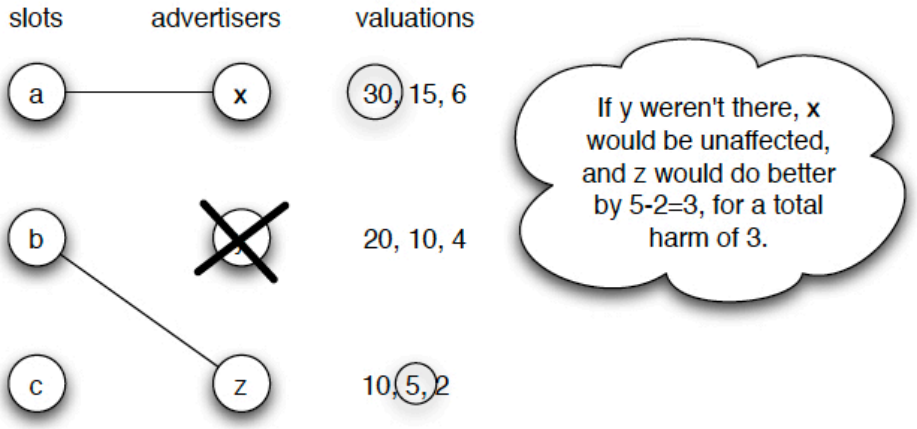
A Sponsored Search Market



The VCG Price in the Market

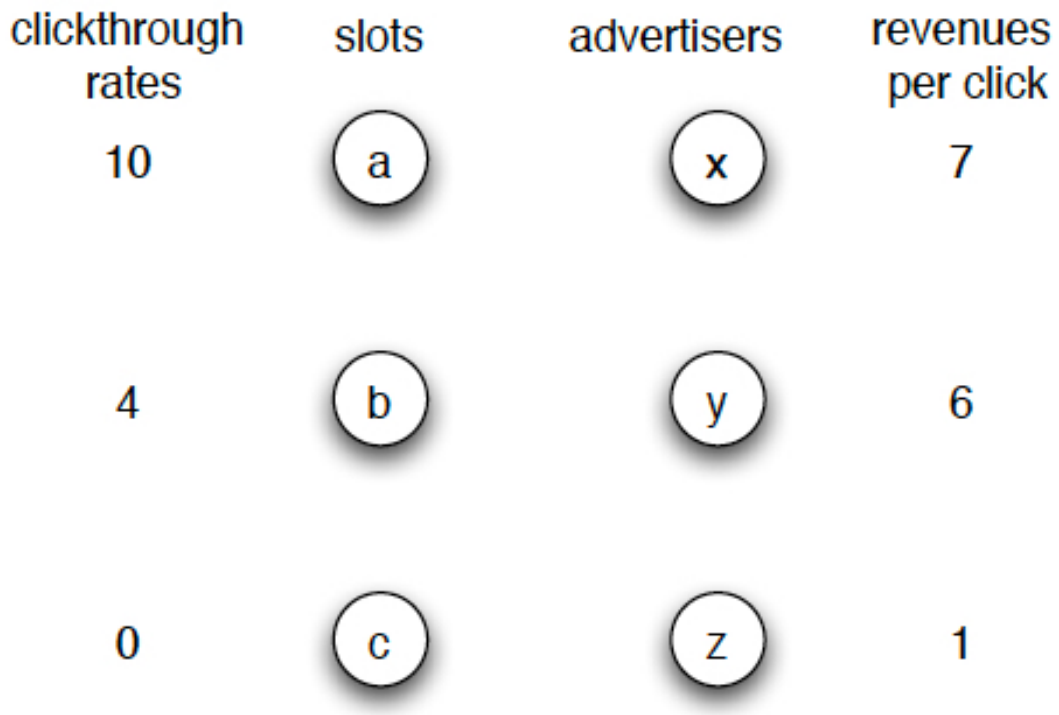


(a) *Determining how much better off y and z would be if x were not present*



(b) *Determining how much better off x and z would be if y were not present*

Another Sponsored Search Market



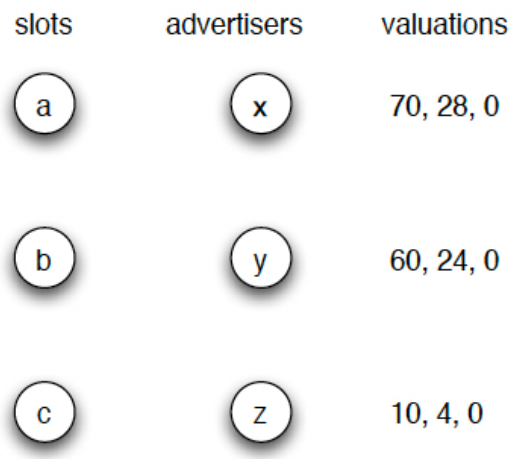
Another Sponsored Search Market: The GSP Auction

Truth-telling may not be an equilibrium For advertiser x, bidding truthfully at 7 gives the payoff of 10, while bidding at 5 gives the payoff of 24.

Multiple and Nonoptimal Equilibria This market has multiple equilibrium

1. If x bids 5, y bids 4, and z bids 2. This forms a Nash equilibrium. This leads to an socially optimal allocation.
2. If x bids 3, y bids 5, and z bids 1. This is also a Nash equilibrium, but not socially optimal.

The Revenue of the GSP and the VCG



Mechanism	Revenue
VCG	44
GSP (Equilibrium 1)	48
GSP (Equilibrium 2)	34

Thank You